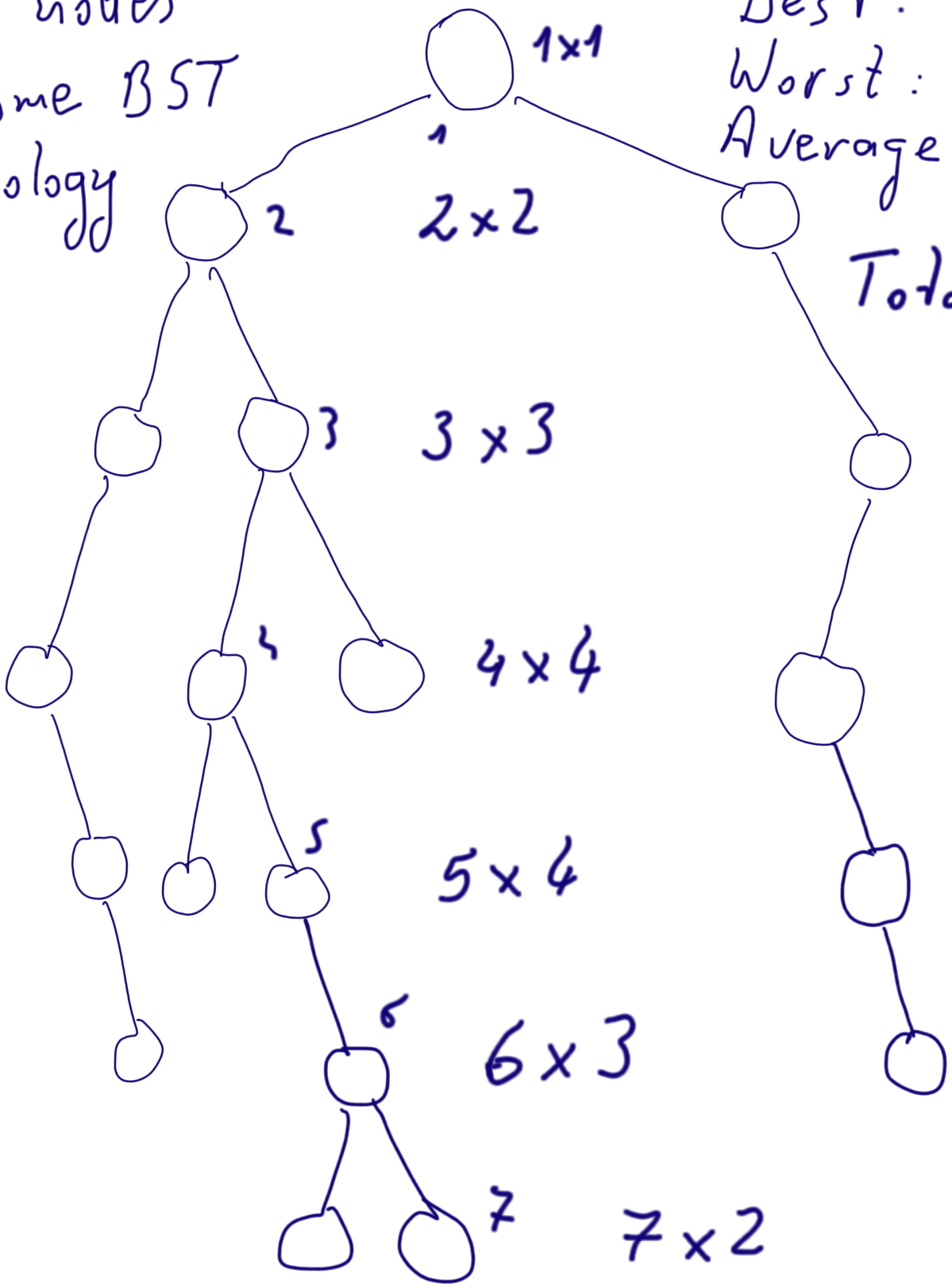
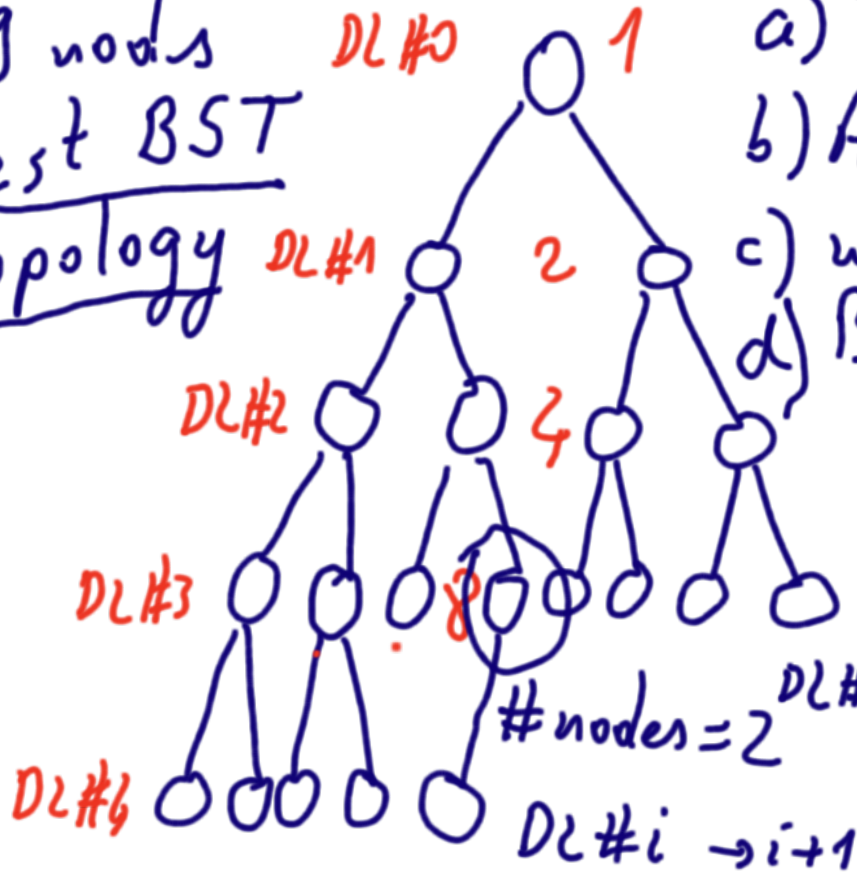
19 nodes
Some BST
topology



Best: 1
Worst: 7
Average: $\frac{82}{19}$
Total: 82

19 nodes

Best BST topology



a) Total access time
b) Average " "
c) Worst " "
d) Best " "

a) 69 "
b) 69/19
c) 5
d) 1

Q) n nodes in a BST with a best distribution of nodes
Provide the distribution of nodes at depth levels?

DL#	0	1	2	3	4	...	i	...	k	x
#of nodes	1	2	4	8	16	...	2^i	...	2^k	?

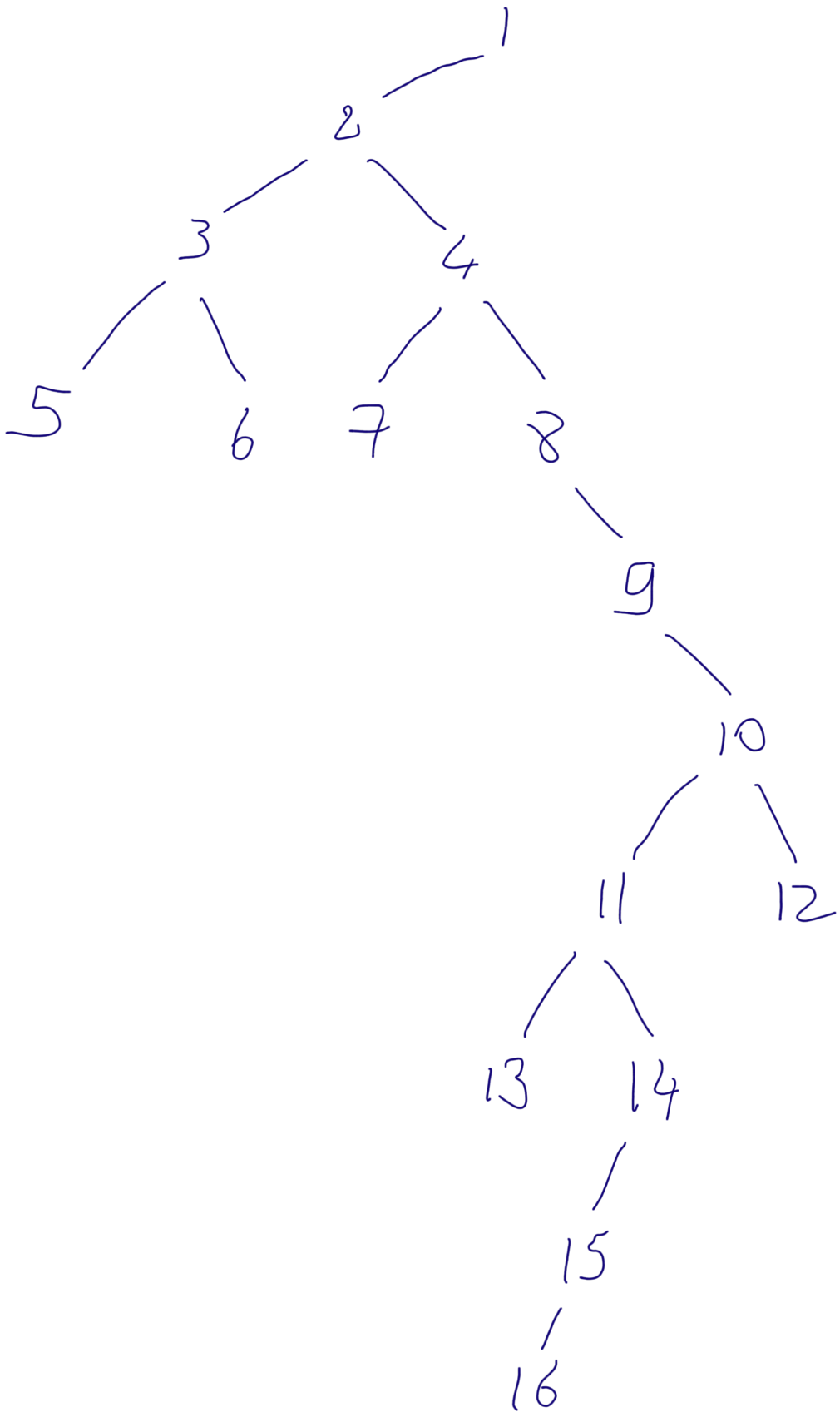
x: # nodes at the last DL

$x = n - (2^{k+1} - 1)$; $k = ?\, f(n)$ in terms of n only

e.g., for $\forall n \in \{16, \dots, 31\}$ $2^{k+1} - 1 = 15$

$\Rightarrow 2^{\lfloor \log_2 n \rfloor} - 1 \Rightarrow k+1 = \lfloor \log_2 n \rfloor$

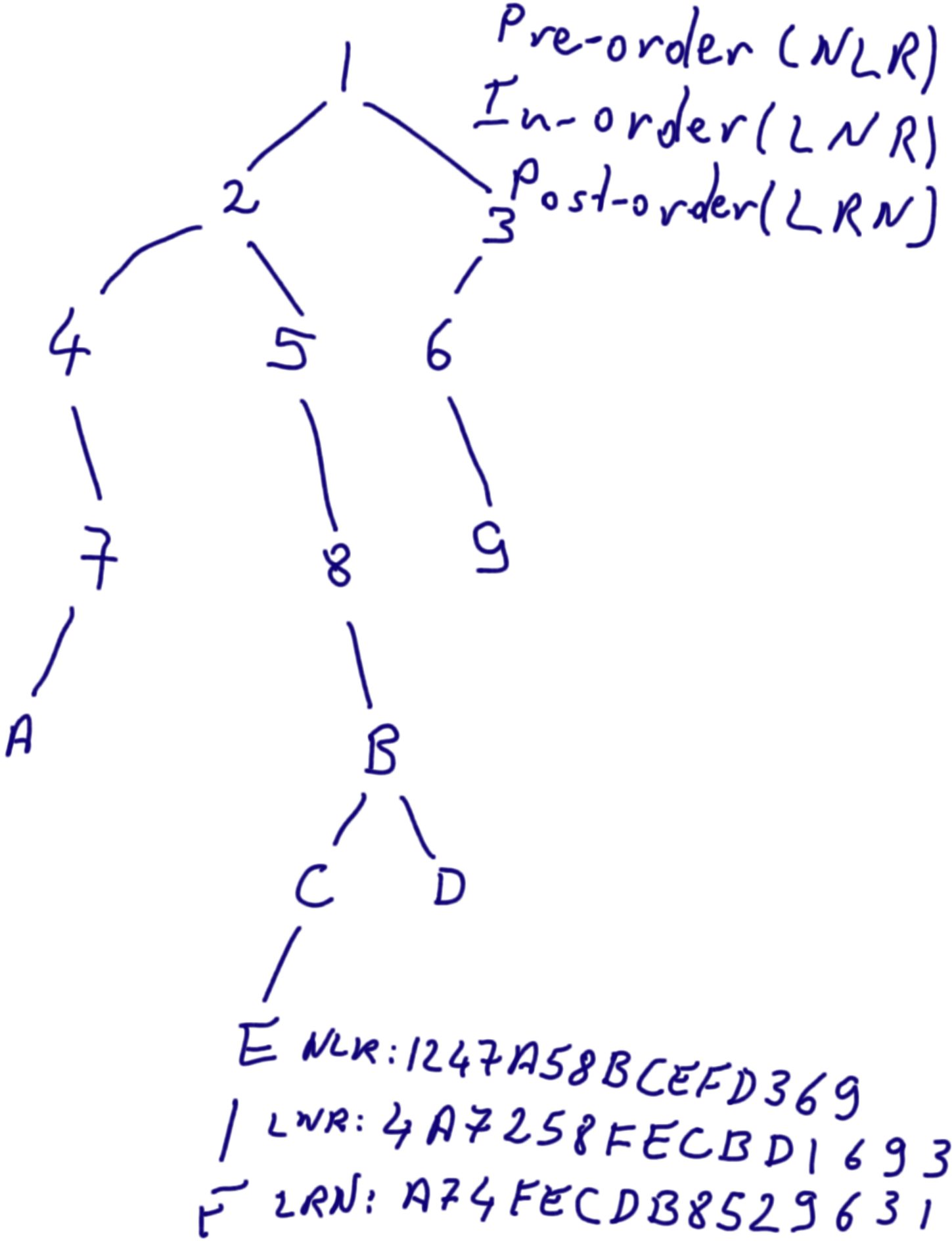
$\Rightarrow x = n - [2^{\lfloor \log_2 n \rfloor} - 1]$



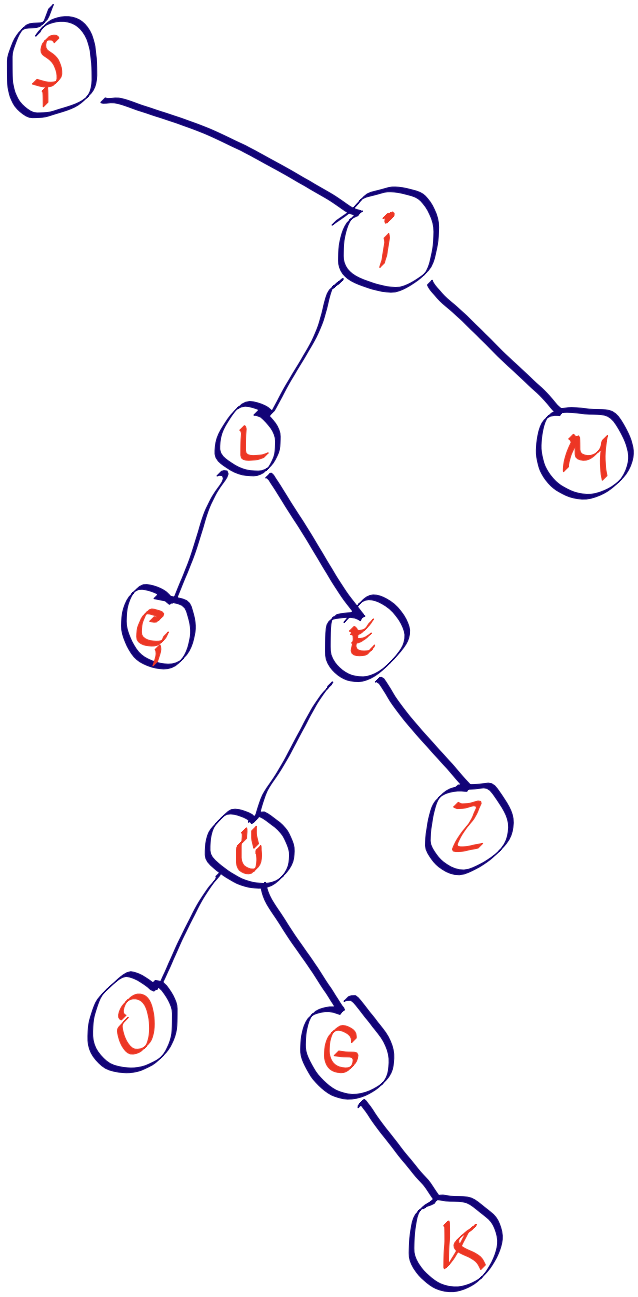
NLR : 1 2 3 5 6 4 7 8 9 10 11 13 14 15 16 12

LNR : 5 3 6 2 7 4 8 9 13 11 16 15 14 10 12 1

LRN : 5 6 3 7 13 16 15 14 11 12 10 9 8 4 2 1

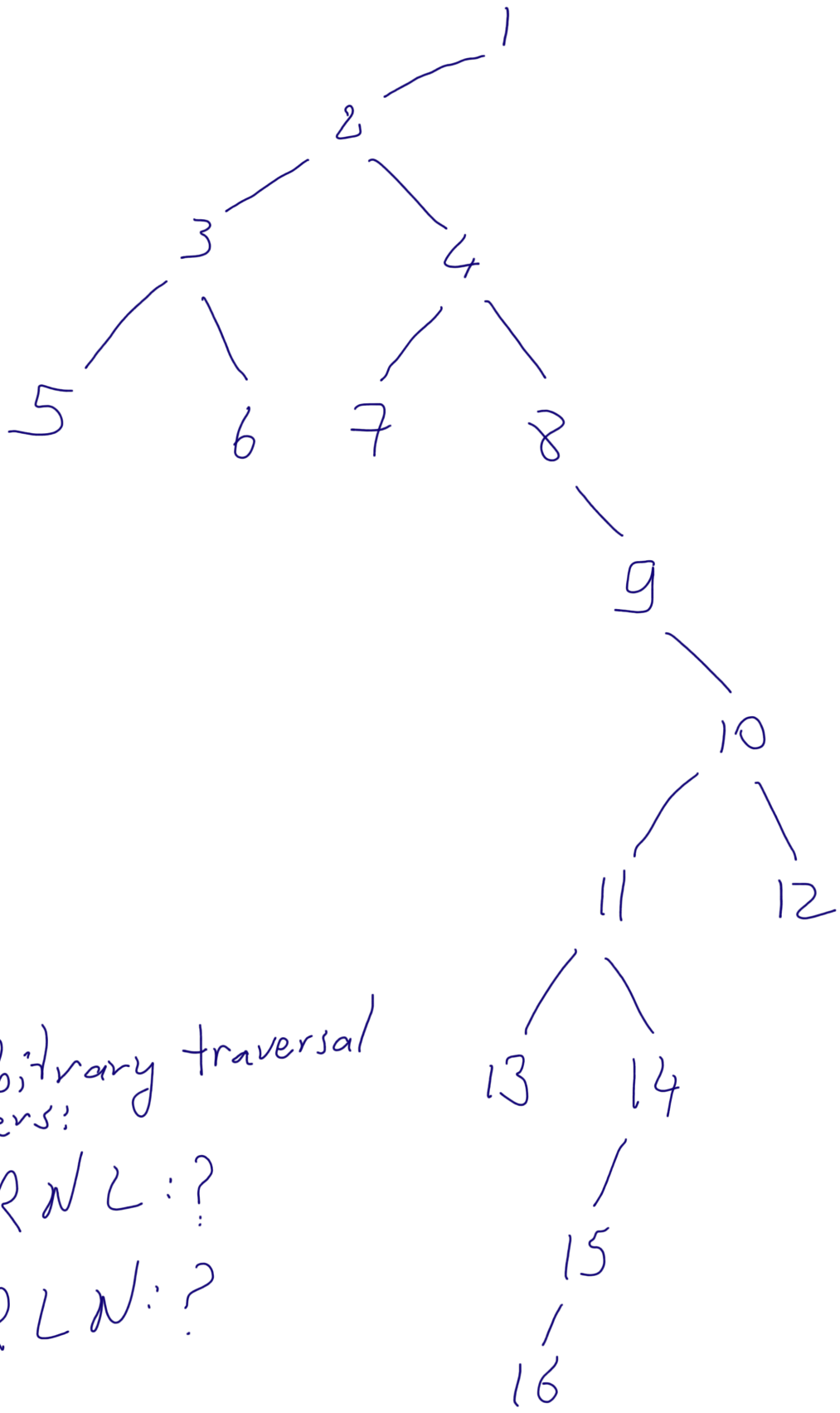

Pre-order (NLR)
In-order (LNR)
Post-order (LRN)
1
2
3
4
5
6
7
8
9
A
B
C
D
E
F
NLR: 1247A58BCEFD369
LNR: 4A7258FECBD1693
LRN: A74FECDB8529631

Place the relevant letters in the BT topology below such that you obtain the string "ÇOK GÜZELMİŞ" as a result of post-order (LRN) traversal!



Arithmetic strings

Infix	Prefix	Postfix
A+B	+AB	AB+
A-B+C or (A-B)+C	-AB+C	AB-C+
A-(B+C)	-A+BC	ABC+-
A*(B+C)/D	/*A+BCD	ABC+*D/
A/(B-C)*(D+E)	*/A-BC+DE	ABC-/DE+*
(A-B)*(C-D/E+F)^(G-H)	*-AB^+-C/DEF-GH	AB-CDE/-F+GH-^*



Arbitrary traversal orders:

RNL: ?

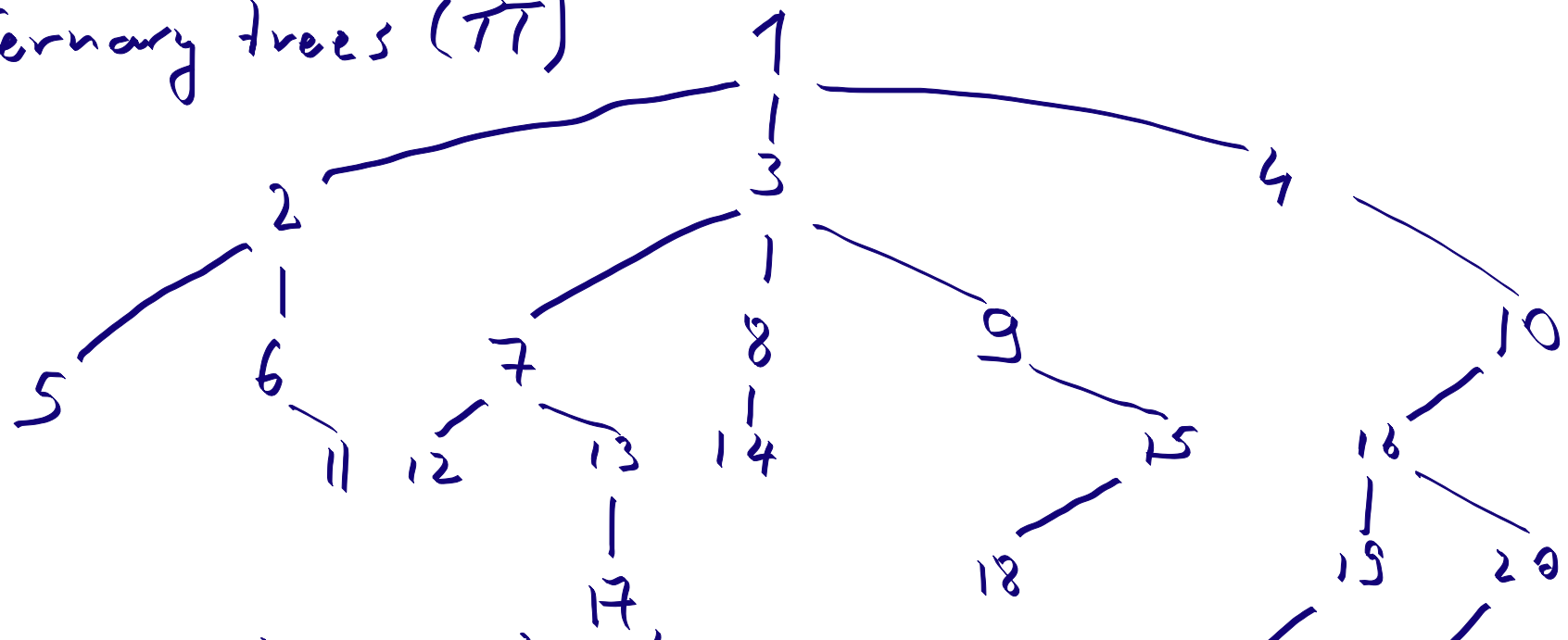
RLN: ?

NLR: 1 2 3 5 6 4 7 8 9 10 11 13 14 15 16 12

LNR: 5 3 6 2 7 4 8 9 13 11 16 15 14 10 12 1

LRN: 5 6 3 7 13 16 15 14 11 12 10 9 8 4 2 1

Ternary trees (TT)



Arbitrary traversal orders:

Left Node Middle Right (LNMR): ?

MLNR: ?

NLMR: 1 2 5 6 11 3 7 12 13 17 8 14 9 15 18 4 10 16 19 21 20 22

NLMR: ?

RLNM: ?

⋮

etc